



P101 · Unit 1 · Quick reference

Physical quantities and executable models · Version 1.1 · 3 October 2026

Physics

Recall · Units	1 km = 1000 m; 1 m = 100 cm = 1000 mm. 1 s = 1000 ms; 1 min = 60 s; 1 kg = 1000 g.
Recall · Changes	$\Delta t = t - t_0$; $\Delta x = x - x_0$
Understand · Model	$x = x_0 + v (t - t_0)$ Fixed origin and direction; constant signed rate v . Use consistent units. Forward-only cases: $t \geq t_0$.
Understand · Checks	$t = t_0$ or $v = 0 \Rightarrow x = x_0$. Test negative rates and nonzero origins. Area: square the length factor; volume: cube it.

Predictions are not measurements; passing code checks does not validate a physical model.

Python · Consult

Assignment and output

```
x_m = 2.5          # metres
x_m = x_m + 1.0
print(x_m)         # 3.5
```

Decision

```
t_s = 8.0
t0_s = 5.0
if t_s >= t0_s:
    print(t_s - t0_s)
else:
    print("Before start")
```

Function and call

```
def position(t, x0, v, t0):
    return x0 + v * (t - t0)

print(position(8.0, 2.5, -0.4, 5.0))
```

Lists (index from 0)

```
values = [2.0, 4.0]
values.append(6.0)
print(values[0]) # 2.0
print(len(values)) # 3
```

Loop and stored results

```
results = []
for t_s in [0.0, 2.0, 4.0]:
    results.append(3.0 * t_s)
print(results) # [0.0, 6.0, 12.0]
```

Comparisons: `==` `!=` `<` `<=` `>` `>=`. Indent blocks consistently.

`return` supplies a result; `print` displays it. No `return` $\rightarrow$ `None`.

Pass dependencies as inputs. After edits: restart and run all cells.